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Week 5: Structure from Motion

- ✔

Ungraded Plugin: 5.1

Overview of Structure from Motion

4 min
- ✔

Practice Assignment: 5.1

Overview of Structure From Motion Self-Check Quiz

Submitted
- ✔

Ungraded Plugin: 5.2

Structure from Motion Problem

4 min
- ✔

Practice Assignment: 5.2

Structure from Motion Problem Self-Check Quiz

Submitted
- ✔

Reading: 5.3 Video Correction

10 min
- ⏪

Ungraded Plugin: 5.3

Observation Matrix

15 min
- 📅

Practice Assignment: 5.3

Observation Matrix Self-Check Quiz

3h
- 📖

Reading: 5.4 Video Correction

10 min
- ⏪

Ungraded Plugin: 5.4

Rank of Observation Matrix

10 min
- 📅

Practice Assignment: 5.4

Rank of Observation Matrix Self-Check Quiz

3h
- ⏪

Ungraded Plugin: 5.5

Tomasi-Kanade Factorization

21 min
- 📅

Practice Assignment: 5.5

Tomasi-Kanade Factorization Self-Check Quiz

3h
- 💬

Discussion Prompt: Week 5

Structure from Motion

10 min
- 💬

Discussion Prompt: Week 5

Questions and Feedback

Week 5 Quiz

Post-Course Survey

5.3 Video Correction

In [Video 5.3 Observation Matrix](#), from 10:40 to 11:13, the equations of $\tilde{u}_{f,p}$ and $\tilde{v}_{f,p}$ should be replaced by $= \mathbf{i}_f^T(P_p - C_f) + (\mathbf{i}_f^T C_f)$ and $= \mathbf{j}_f^T(P_p - C_f) + (\mathbf{j}_f^T C_f)$.

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Centering Trick

3D centroid

Scene

Frame f

2D centroid $(\bar{u}_f, \bar{v}_f)$

Shift camera origin to the centroid $(\bar{u}_f, \bar{v}_f)$.

Image points w.r.t. $(\bar{u}_f, \bar{v}_f)$:

$$\begin{aligned} \tilde{u}_{f,p} &= u_{f,p} - \bar{u}_f \\ &= \mathbf{i}_f^T(P_p - C_f) + \mathbf{i}_f^T C_f \\ \tilde{u}_{f,p} &= \mathbf{i}_f^T P_p \end{aligned}$$

$$\begin{aligned} \tilde{v}_{f,p} &= v_{f,p} - \bar{v}_f \\ &= \mathbf{j}_f^T(P_p - C_f) + \mathbf{j}_f^T C_f \\ \tilde{v}_{f,p} &= \mathbf{j}_f^T P_p \end{aligned}$$

Camera locations C_f now removed from equations.

From 12:05 to 14:08, point 1 and point 2 in the penultimate row of the matrix form in the slide should be $\tilde{v}_{2,1}$ and $\tilde{v}_{2,2}$ instead of $\tilde{u}_{2,1}$ and $\tilde{u}_{2,2}$.

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Observation Matrix W

$$\begin{aligned} \tilde{u}_{f,p} &= \mathbf{i}_f^T P_p \\ \tilde{v}_{f,p} &= \mathbf{j}_f^T P_p \end{aligned} \Rightarrow \begin{bmatrix} \tilde{u}_{f,p} \\ \tilde{v}_{f,p} \end{bmatrix} = \begin{bmatrix} \mathbf{i}_f^T \\ \mathbf{j}_f^T \end{bmatrix} P_p$$

	Point 1	Point 2	...	Point N
Image 1	$\tilde{u}_{1,1}$	$\tilde{u}_{1,2}$	...	$\tilde{u}_{1,N}$
Image 2	$\tilde{u}_{2,1}$	$\tilde{u}_{2,2}$	...	$\tilde{u}_{2,N}$
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Image F	$\tilde{u}_{F,1}$	$\tilde{u}_{F,2}$	...	$\tilde{u}_{F,N}$

$W_{2F \times N}$
Centroid-Subtracted Feature Points (Known)

$$\begin{bmatrix} \tilde{u}_{f,p} \\ \tilde{v}_{f,p} \end{bmatrix} = \begin{bmatrix} \mathbf{i}_f^T \\ \mathbf{j}_f^T \end{bmatrix} P_p$$

$M_{2F \times 3}$
Camera Motion (Unknown)

	Point 1	Point 2	...	Point N
Image 1	$\tilde{v}_{1,1}$	$\tilde{v}_{1,2}$	...	$\tilde{v}_{1,N}$
Image 2	$\tilde{v}_{2,1}$	$\tilde{v}_{2,2}$	...	$\tilde{v}_{2,N}$
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Image F	$\tilde{v}_{F,1}$	$\tilde{v}_{F,2}$	...	$\tilde{v}_{F,N}$

$S_{3 \times N}$
Scene Structure (Unknown)

$[P_1 \ P_2 \ \dots \ P_N]$

From 14:09 to 14:40, point 1 and point 2 in the penultimate row of the matrix form in the slide should be $\tilde{v}_{2,1}$ and $\tilde{v}_{2,2}$ instead of $\tilde{u}_{2,1}$ and $\tilde{u}_{2,2}$.

*The revisions are shown in **red** and correspond to either corrections to errors or additions made for clarity.

Observation Matrix W

	Point 1	Point 2	...	Point N
Image 1	$\tilde{u}_{1,1}$	$\tilde{u}_{1,2}$	...	$\tilde{u}_{1,N}$
Image 2	$\tilde{u}_{2,1}$	$\tilde{u}_{2,2}$	...	$\tilde{u}_{2,N}$
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Image F	$\tilde{u}_{F,1}$	$\tilde{u}_{F,2}$	...	$\tilde{u}_{F,N}$

$W_{2F \times N}$
Centroid-Subtracted Feature Points (Known)

$$\begin{bmatrix} \tilde{u}_{f,p} \\ \tilde{v}_{f,p} \end{bmatrix} = \begin{bmatrix} \mathbf{i}_f^T \\ \mathbf{j}_f^T \end{bmatrix} P_p$$

$M_{2F \times 3}$
Camera Motion (Unknown)

	Point 1	Point 2	...	Point N
Image 1	$\tilde{v}_{1,1}$	$\tilde{v}_{1,2}$	...	$\tilde{v}_{1,N}$
Image 2	$\tilde{v}_{2,1}$	$\tilde{v}_{2,2}$	...	$\tilde{v}_{2,N}$
...	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Image F	$\tilde{v}_{F,1}$	$\tilde{v}_{F,2}$	...	$\tilde{v}_{F,N}$

$S_{3 \times N}$
Scene Structure (Unknown)

$[P_1 \ P_2 \ \dots \ P_N]$

Can we find M and S from W ?

Go to next item

✔ Completed

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